INCH-POUND
MIL-PRF-38201D
w/AMENDMENT 1
03 JULY 2025
SUPERSEDING
MIL-PRF-38201D
18 DEC 1998

PERFORMANCE SPECIFICATION
VALVE, FILLER, LIQUID OXYGEN, FEMALE, CRU-59/E

Reactivated after (3 July 2925) and may be used for new and existing designs and acquisitions.



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1. SCOPE

1.1 Scope. This specification covers one type of CRU-59/E female liquid oxygen filler valve.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in Sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in Sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25961	Valve, Fill-Buildup-Vent, Liquid Oxygen Converter, CRU-50/A
MIL-PRF-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-HDBK-781	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	Nitrogen, Technical
A-A-59155	Nitrogen, High Purity, Special Purpose

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD-adopted)

(Copies of these documents are available online at https://asq.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

3.3 Materials. All materials shall be suitably treated to resist corrosion due to electrolytic decomposition and any other atmospheric conditions that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals, (ODCs) shall be avoided, whenever feasible.

3.3.1 Aluminum and aluminum alloys. Aluminum and aluminum alloys, if used, shall be anodized (see 6.4) or have an alternate coating with equivalent characteristics (see 6.2).

3.4 Design. The CRU-59/E valve shall provide a positive leak-tight connection with a CRU-50/A valve conforming to MIL-PRF-25961 for use in filling a liquid oxygen converter from a liquid oxygen charging cart. The valve shall contain a poppet (see 6.5) that is normally seated, preventing a flow through the valve, and shall be automatically unseated when two valves are completely connected. Complete connection of the two valves shall require a two-phase operation. The valves shall be connected mechanically without flowing through the valves in the primary phase. The secondary phase shall incorporate suitable actuation by one of the valves, to allow the flow of liquid oxygen. Disconnection shall not be necessary to cease flow. Suitable means, such as reverse rotation, shall be incorporated to stop flow. If rotation is used, the amount required to actuate or cease flow shall not exceed 100 degrees. When the two valves are completely disconnected, the valve shall automatically return to its neutral position ready for reconnection. The valve used for interface and testing of the CRU-59E valve shall be the CRU-50/A valve conforming to MIL-PRF-25961 previously qualified under the applicable Qualified Products List.

3.4.1 Valve sealing. The mating of the valves shall form an effective leaktight connection which shall resist inadvertent disconnection due to the minor shocks and jars associated with filling operations.

3.4.2 Internal seals. The internal seals of the valve assembly shall be designed in such a manner that the entrapment of liquid oxygen cannot take place between screw threads, if used, and sealing surfaces.

3.4.3 Closures. All openings of the valve shall be closeable with plugs, caps, or similar devices that shall prevent dust and any foreign matter from entering the valve when not in use.

3.4.3.1 Valve cap. A cap shall be provided for the outlet end of the valve. The valve shall have an external gripping surface. The cap shall be attached to the valve housing while allowing installation and removal of the cap. The attachment shall have a minimum breaking strength of 130 pounds, shall be non-sparking, and shall turn freely to minimize kinking.

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3.5 Orientation. The valve shall operate while oriented in any position.

3.6 Interface. The valve shall be as defined by figure 1 and shall conform to the envelope dimensions shown in figure 2.

3.6.1 Screw threads. Screw threads, if used on the valve, shall conform to MIL-S-7742 or any other commercially acceptable thread standard. If screw threads are used in assembling the valve they shall be securely locked by lockwashers, locknuts, or any other low temperature oxygen compatible means.

3.6.2 Weight. The total weight of the valve assembly shall not exceed three pounds (lbs).

3.6.3 Engagement pins. The valve designs shall incorporate engagement pins similar to those shown on figure 2. The pins shall interface with the CRU-50/A valve assembly.

3.7 Performance.

3.7.1 Leakage (CRU-59/E valve only). Leakage of the valve shall not exceed 0.05 liters per minute (lpm) when the valve is disconnected and a gaseous pressure of 50 lbs per square inch gauge (psig) is applied to the inlet connection.

3.7.2 Leakage (primary connection). Combined leakage of the valves shall not exceed 0.05 lpm when the valve is primarily connected and a gaseous pressure of 50 lbs psig is applied to the inlet connection.

3.7.3 Leakage (complete connection, liquid). There shall be no visual evidence of leakage from a completely connected valve when a flow of test liquid is applied to the CRU-59/E valve inlet under a pressure of 30 psig.

3.7.3.1 Leakage (complete connection, gaseous). Leakage from a completely connected valve shall not exceed 0.05 lpm when subjected to a gaseous pressure of 50 psig applied to the inlet connection of the CRU-59/E valve.

3.7.4 Flow rate. Flow rate from the CRU-50/A valve shall be greater than 40 lpm when a maximum pressure of 3 psig is applied to the inlet of the CRU-59/E valve.

3.7.5 Simulated aircraft-mounted converter filling. There shall be no leakage from a completely connected valve mounted in a horizontal or vertical position when a flow of test liquid under a minimum pressure of 30 psig at the valve inlet is passed through both valves.

3.7.6 Connection, disconnection torque. The torque required to make the connection and disconnection of the CRU-59/E and CRU-50/A valves shall be between 20 and 85-inch pounds.

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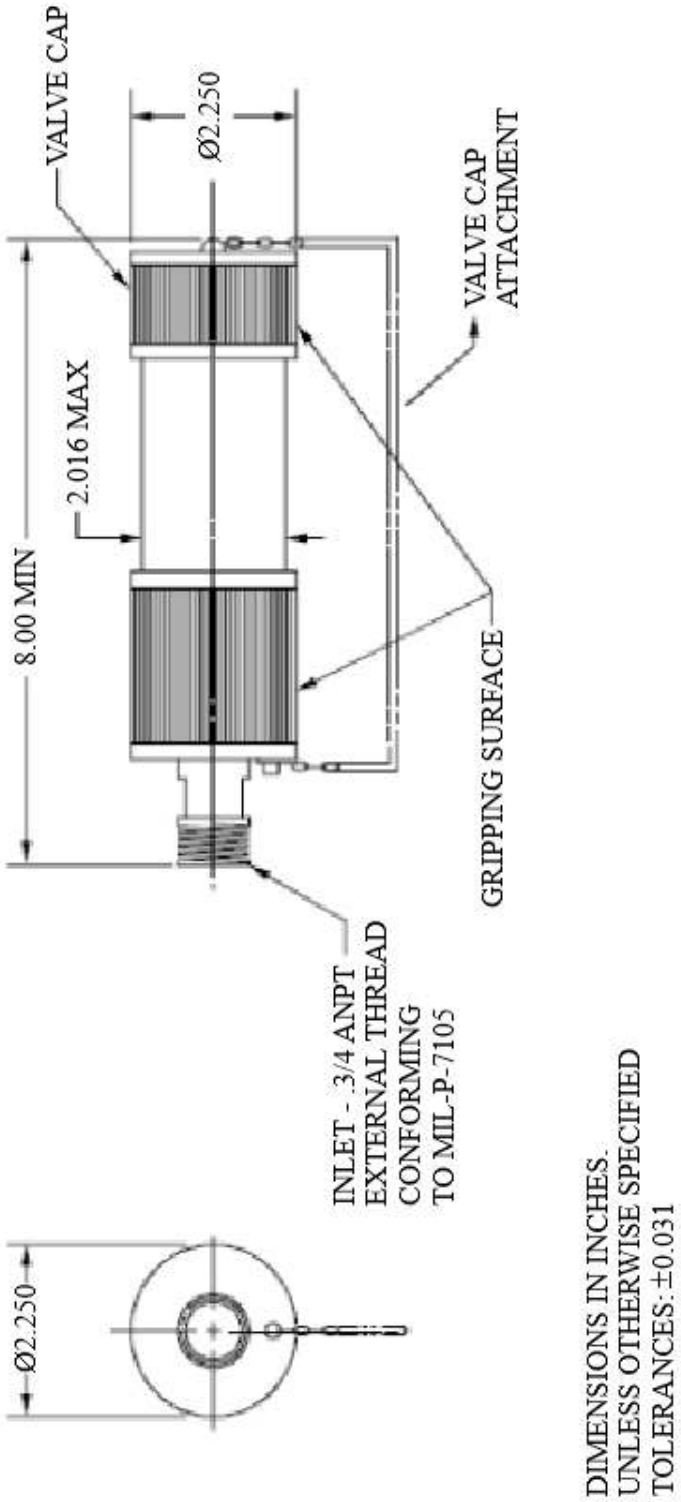


FIGURE 1. Valve.

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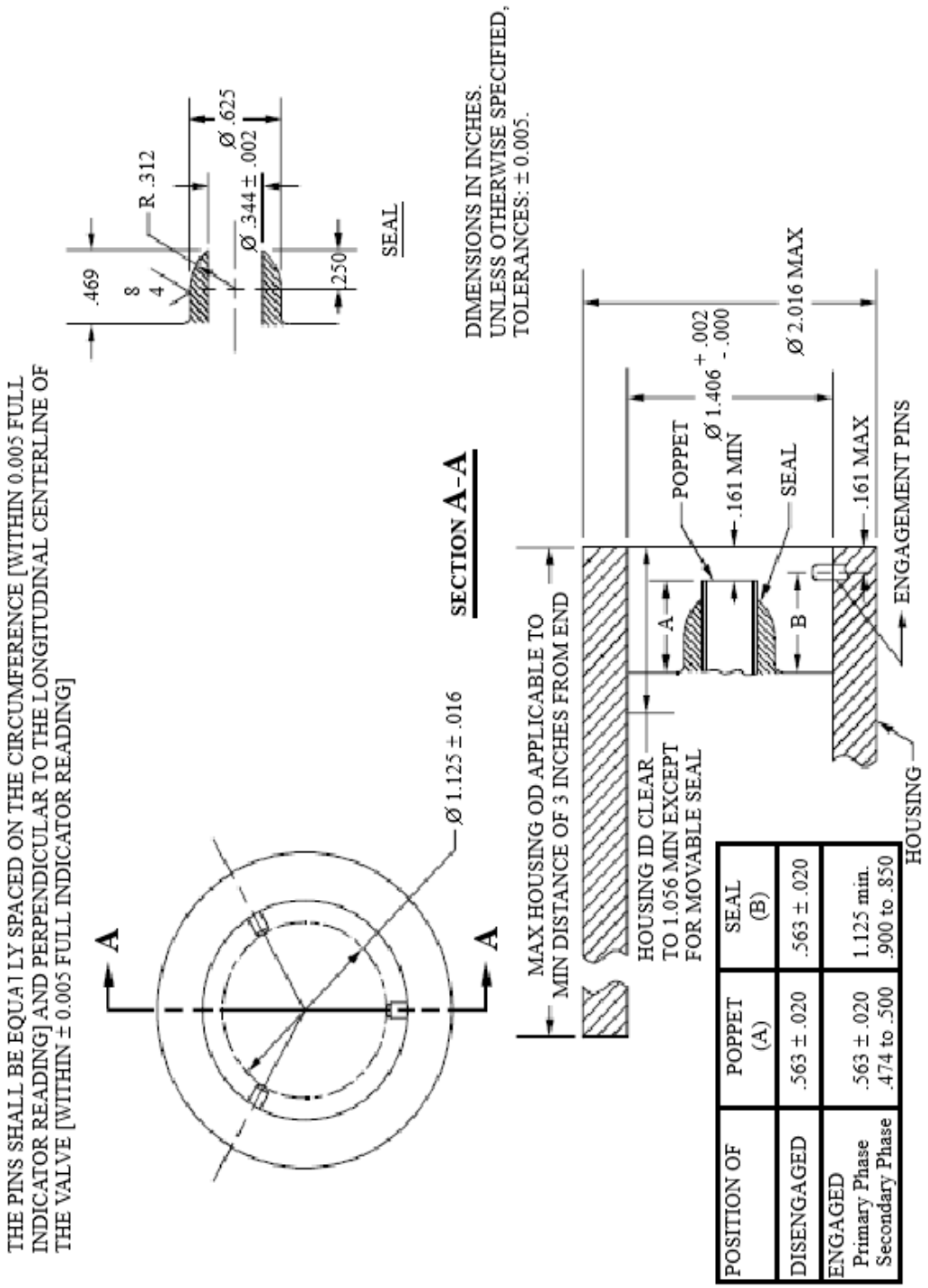


FIGURE 2. Female fill valve seal.

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3.8 Environmental conditions. The valve shall operate and not show evidence of damage, such as cracks, permanent corrosion, deformation, leaks, or other flaws, during or after exposure to the following conditions:

a. Exposure to temperatures ranging from –65 to +160 °F.

b. Vibration incident to service use.

c. Shock incident to service use.

3.9 Reliability. The valve shall have a specified mean cycles between failures (MCBF) of not less than 700 cycles at the 90 percent lower confidence limit.

3.10 Item identification. A nameplate, permanently and legibly marked, shall be securely attached or stamped in a conspicuous place on the valve, but shall not interfere with the operation of the valve. The information required is as follows:

Nomenclature

Applicable Military Specification

Design Activity, CAGE Code

Manufacturer's I.D. (If different from design activity)

Contract or Purchase Order

Serial number (if assigned)

The acquisition documents may require additional information.

3.10.1 Warning label. A white, non-conductive label with a red-lettered statement shall be permanently affixed to the valve and shall contain the following information:

CAUTION

USE TWO-STEP PROCEDURE

3.11 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.12 Cleaning. See MIL-STD-1330 and MIL-STD-1359 for guidance on proven cleaning methods and verifications. Visible inspections are typically conducted using white light and ultraviolet light to detect particulate and some types of hydrocarbons. The non-volatile residue (NVR) test in MIL-STD-1359 may be used to baseline the hydrocarbon verification at an acceptable contamination level; however, other cleaning and verification methods that do not contain class I and II ODC solvents should be used for production.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified by the procuring activity (see 6.2), one valve assembly shall be subjected to all the inspections and tests in 4.6 and 4.7.

4.2.1 Reliability test. One preproduction sample, after passing all other inspections, shall be tested for reliability as specified in 4.9.

4.3 Conformance inspection. Conformance inspections shall include tests in 4.3.1 and 4.3.2.

4.3.1 Individual tests. Each valve shall be subjected to the following tests and examination:

a. Examination of product (see 4.6).

b. Leakage (CRU-59/E) (see 4.7.1).

4.3.2 Sampling inspection. Samples of liquid oxygen valves shall be selected from a lot of completed inflation devices in accordance with inspection Level II of ANSI/ASQC Z1.4, or equivalent, and subjected to the following tests:

a. Leakage (primary connection) (see 4.7.2).

b. Leakage (complete connection) (see 4.7.3).

c. Flow rate (see 4.7.4).

d. Simulated aircraft-mounted converter filling (see 4.7.5).

e. Connection, disconnection torque (see 4.7.6).

f. Reliability (see 4.9).

4.4 Requirements cross-reference matrix. Table I provides a cross-reference matrix of the Section 3 requirements tested or verified in the paragraphs below.

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TABLE I. Requirements cross-reference matrix.

REQUIREMENT	VERIFICATION	REQUIREMENT	VERIFICATION
3.2	4.6	3.7.1	4.7.1
3.3	4.6	3.7.2	4.7.2
3.3.1	4.6	3.7.3	4.7.3
3.4	4.6	3.7.4	4.7.4
3.4.1	4.7.3	3.7.5	4.7.5
3.4.2	4.7.5	3.7.6	4.7.6
3.4.3	4.6	3.8a	4.8.1, 4.8.2
3.4.3.1	4.6	3.8b	4.8.3
3.5	4.7.5	3.8c	4.8.4
3.6	4.6	3.9	4.9
3.6.1	4.6	3.10	4.6
3.6.2	4.6	3.11	4.6
3.6.3	4.6		

4.5 Test conditions. Unless otherwise specified, all inspections shall be performed under ambient conditions, i.e., atmospheric pressure of 28 to 32-inches of mercury, temperature of 59 to 95 °F, and relative humidity of not more than 80%. If conditions change during a test, the test shall be suspended and appropriate corrections applied before resuming the test. Unless otherwise indicated in the test procedures, test sample temperature conditioning shall be at least one hour longer than that required for the sample to reach thermal equilibrium.

4.5.1 Test medium. Unless otherwise specified, the test medium shall be oxygen conforming to MIL-PRF-27210, liquid and gas, or nitrogen conforming to A-A-59503 and A-A-59155 liquid and gas, Class 1, Grade B. The units shall be purged prior to each phase of the test program using gaseous nitrogen conforming to A-A-59503 and A-A-59155 (Type 1) Class 1, Grade B.

4.6 Examination of product. Each valve shall be carefully examined to determine conformance with respect to materials, interface, design, identification, and interchangeability.

4.7 Tests.

4.7.1 Leakage (CRU-59/E valve only). With the valve, tested as specified in 4.7.3." disconnected, a gaseous pressure of 50 psig shall be applied to the inlet connection for at least 1 minute. The gas leakage from the valve shall not exceed 0.05 lpm. The poppet (see 6.5) shall be unseated 3 times prior to this test.

4.7.2 Leakage (primary connection). With the valve mechanically connected to a valve conforming to MIL-PRF-25961 and with the mechanism that unseats the poppets in the seated position, a gaseous pressure of 50 psig shall be applied to the inlet connection of the CRU-59/E valve for at least 1 minute. With the pressure applied, the combined leakage from the CRU-59/E valve and the outlet of the valve conforming to MIL-PRF-25961 shall not exceed 0.05 lpm. The poppet seal shall be unseated 3 times prior to this test.

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4.7.3 Leakage (complete connection). The CRU-59/E valve shall be attached to a liquid oxygen supply source at 30 psig. A CRU-50/A valve (MIL-PRF-25961) shall be fitted to a 300 psig working pressure LOX converter in a simulated aircraft manner. The converter shall be partially filled and shall be at operating pressure. With the valve completely connected (poppets unseated) to the CRU-50/A valve. A flow of test liquid under a pressure of 30 psig measured at the valve inlet shall be passed through the two valves for at least 10 minutes. During this test, the valves shall be disconnected and reconnected at least three times. Under all conditions of test liquid flow and disconnections, there shall be no visual evidence of leakage of test liquid from either the poppet or interior areas, as evidenced by any liquid flow. The CRU-59/E valve shall then be subjected to the tests of paragraph 4.7.1.

4.7.4 Flow rate. The valve shall be completely connected (poppets unseated) to a CRU-59/A valve and a flowmeter shall be connected to the outlet of the valve by means of a 0.375-inch tube. A maximum pressure of 3 psig shall be applied to the inlet of the CRU-59/E valve and flow from the CRU-59/A valve shall be greater than 40 lpm.

4.7.5 Simulated aircraft-mounted converter filling. The valve shall be connected to a valve conforming to MIL-PRF-25961 that is mounted in a horizontal position as shown in figure 3. A flow of test liquid under a minimum pressure of 30 psig at the valve inlet shall be passed through both valves. There shall be no evidence of test liquid leakage from the valve or from the mating surfaces. This test shall be repeated with the MIL-PRF-25961 valve mounted in a vertical position with the inlet of the CRU-59/E valve down. During both of these tests the valves shall be disconnected and visually inspected for entrapment of liquid between mating surfaces and leakage.

4.7.6 Connection, disconnection torque. The CRU-59/E valve shall be completely connected (poppets unseated) to a CRU-59A valve and then completely disconnected. The torque required to make this connection and disconnection shall not exceed 85 inch-pounds nor be less than 20 inch-pounds.

4.8 Environmental tests.

4.8.1 High temperature exposure. The valve shall be exposed to a temperature of 160 ± 5 °F for a minimum of four hours. While exposed to this temperature, the valve assembly shall be subjected to and pass the examination and tests of paragraphs 4.7.1, 4.7.2, and 4.7.3. The valve assembly shall be removed from the chamber and immediately subjected to the test in paragraph 4.7.5.

4.8.2 Low temperature exposure. The valve shall be exposed to a temperature of -65 ± 5 °F for a minimum of four hours. While exposed to this temperature, the valve assembly shall be subjected to and pass the examination and tests of paragraphs 4.7.1, 4.7.2, and 4.7.3. The valve assembly shall be removed from the chamber and immediately subjected to the test in paragraph 4.7.5.

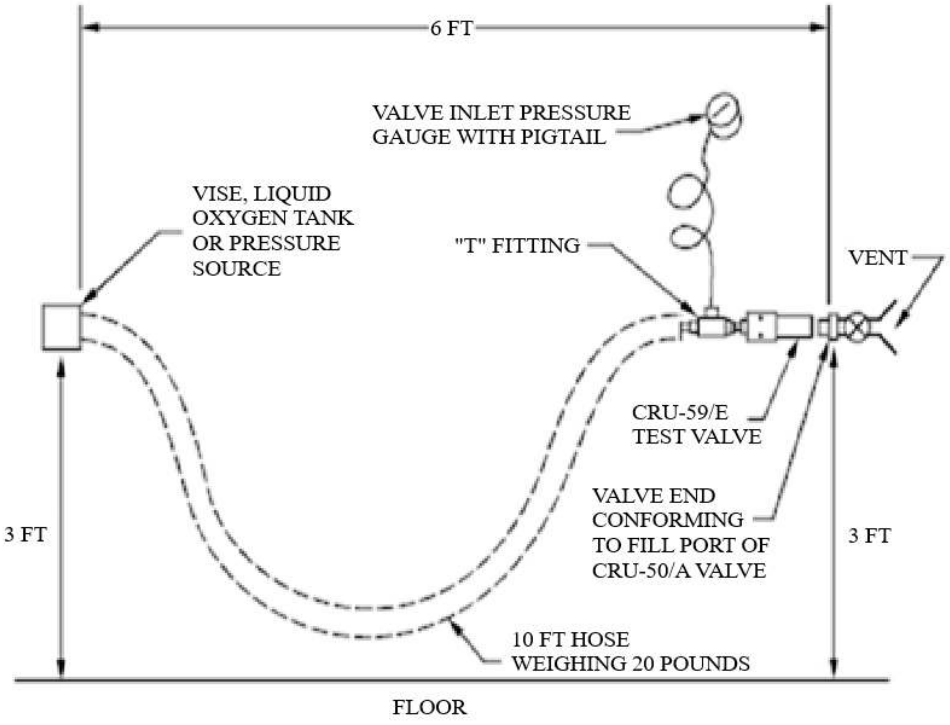
4.8.3 Vibration. The valve assembly with cap shall be tested in accordance with MIL-STD-810, Method 514, test procedure I, category 10. At the conclusion of these tests, the tests in paragraphs 4.7.1, 4.7.2, 4.7.3, 4.7.4, 4.7.5, and 4.7.6 shall be performed.

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4.8.4 Shock. The valve assembly with cap shall be tested in accordance with MIL-STD-810, Method 516, procedure I. At the conclusion of these tests, the tests in paragraphs 4.7.1, 4.7.2, 4.7.3, 4.7.4, 4.7.5 and 4.7.6 shall be performed.

4.9 Reliability. The valve shall be subjected to cycles of connection and disconnection to a mating valve conforming to MIL-PRF-25961, using the test setup similar to the one shown in figure 3. A cycle shall consist of complete connection (poppet unseated) including: venting of the CRU-50/A valve from 300 psig, flow of test liquid through the mated valves, and disconnection (valves separated). The assembly shall be subjected to the tests specified in paragraphs 4.7.1, 4.7.3, 4.7.5, and 4.7.6 every 700 cycles during the reliability test, to determine if a failure has occurred. A valve assembly failing any of these tests shall have failed the reliability test. In addition, any failure of the valve to mate properly or any fracture of components shall constitute a test failure. No parts shall be replaced during the reliability testing as preventive maintenance. Acceptance shall be based on the accept-reject criteria of test plan I of MIL-HDBK-781, as well as recording, data handling, and reporting procedures.

4.10 Pressure Actuation. A CRU-50/A valve shall have the buildup and gas ports connected to a 450 psig gaseous pressure source. The vent and fill outlet ports shall be open. Upon connection with the CRU-59/E valve, the CRU-50/A shall vent the 450 psig pressure.



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FIGURE 3. Simulated aircraft-mounted converter filling.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The valve is intended to be used with a storage and servicing liquid oxygen tank to fill aircraft-mounted converters with liquid oxygen.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. When first article inspection is required (see 3.1).

d. The requirement for the vendor to identify proposed coatings if aluminum or aluminum alloys are used and anodization is not to be used (see 3.3.1).

e. Item identification (see 3.10).

f. Packaging requirements (see 5.1).

g. Data required.

6.3 Anodizing. MIL-A-8625, Type I or II has been used successfully in previous procurements as a guide for anodizing aluminum and aluminum alloys, if used.

6.4 Valve poppet. The term "poppet" in this specification is defined as the valve opening and closing (or sealing) mechanism and is not to be construed to mean only the spring-loaded poppet type of valve. Tests requiring unseating or seating of the poppet should also be interpreted to mean opening or closing of the sealing mechanism, if the valve is not of the traditional spring-loaded poppet type of valve.

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6.5 Environmental testing. Test methods are specified in MIL-STD-810E.

6.6 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

6.7. Subject term (key word) listing.

Valve assembly

Valve cap

Liquid oxygen

Charging cart

LOX

Custodians:
Army - GL
Navy - SA
Air Force - 184

Review activities
Air Force - 11, 71
DLA - GS

Preparing Activity:
Air Force - 184

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